Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Starting position

Meaning in my words How would you explain starting position to a friend?	**How can you tell?** At 0 s the cart is beside the +5 m mark. Where does its position graph begin?
Example or sketch The cart starts at +5 m and moves 2 m left. What two positions would you mark?	**Non-example and why** “The clock starts at zero, so the cart starts at zero.” What do the two zeros mean?

Displacement

Meaning in my words How would you explain displacement to a friend?	**How can you tell?** Which two marks tell you the displacement even if the cart turns around?
Example or sketch Go from 2 m to 6 m and back to 4 m. What single arrow shows the displacement?	**Non-example and why** Someone adds both legs and gets 6 m. What did they find?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Final position

Meaning in my words	How can you tell?
How would you explain final position to a friend?	What must you combine with the starting position to locate the finish?
Example or sketch	**Non-example and why**
A cart starts at +5 m and moves 7 m left. Where would you draw it?	A partner reports 7 m because that is how far the cart traveled. What is missing?

Prediction range

Meaning in my words	How can you tell?
How would you explain prediction range to a friend?	A forecast reads 10 ± 2 km. Which positions are at its ends?
Example or sketch	**Non-example and why**
Where would you shade the allowed positions on a number line?	Someone circles only 10 km. What information disappears?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Mark the central forecast and shade its interval. Label one reason the interval could grow.

Choose zero, direction, and scale.



My marks mean

On assignment: Prediction range

SECTION E / BACK / FINAL MEETING ONLY Name _____

Two XJ teams predict a center of 20 km. Team A gives ±1 km; team B gives ±4 km. A telescope has one narrow field.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Survey engineer**

Draw both possible bands on one axis. Recommend which report requires a wider search; give both sets of endpoints.

☐ **Weather-desk presenter**

Produce a forecast graphic with two bands. Explain to a viewer which search is wider and what the graphic cannot tell us about impact.

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.